4.3 The Complex Plane

Name _____ Period ____ Date _____

Pre-Calculus 4.3 · Student sheet

I can…

- plot complex numbers in the complex plane.
- find the absolute value of a complex number.
- find the distance and midpoint between two complex numbers.

Vocabulary · fill in the missing word

Complex plane — a _____ plane with a horizontal real axis and a vertical imaginary axis

Absolute value of a complex number — $|a + bi| = \sqrt{(a^2 + b^2)}$, its _____ from the origin

Part 1 · Complex numbers as points

Plot 3 + i and 1 + 4i on a grid. Draw arrows from the origin and complete the parallelogram.

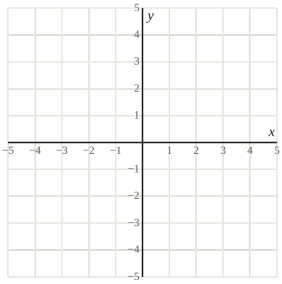
Fourth corner	(3 + i) + (1 + 4i)	Conjugate of 3 + i	How the conjugate moved

Part 2 · Notes and examples (fill in each note, then work the example)

Plotting: a + bi is the point _____: real part across, imaginary part up.

Example 1

Plot 2 + 3i, −1 + 2i, 4 − 2i, and −3i in the complex plane.



Absolute value: |a + bi| = _____

Example 2

Find each absolute value.

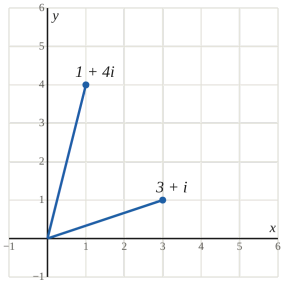
a. |−2 + 5i|

b. |6 − 8i|

Adding complex numbers works like adding _____: the parallelogram rule.

Example 3

Add (3 + i) + (1 + 4i) graphically and check algebraically.



Example 4

Plot 3 + 2i and its conjugate. Describe how they are related.

Distance and midpoint: the distance between z_1 and z_2 is $|z_1 - z_2|$; the midpoint averages the _____.

Example 5

Find the distance and the midpoint between 2 + 3i and 5 − i.

Part 3 · Partner practice

Partner 1

Find $|-5 - 12i|$.

Partner 2

Find the distance and midpoint between −1 + 2i and 3 + 5i.